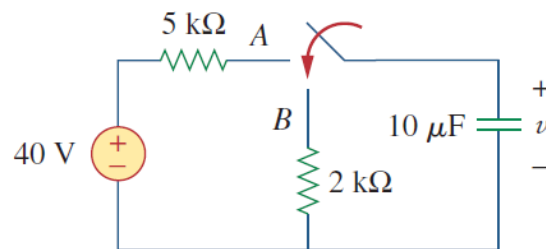


Answer all the question with detail steps.

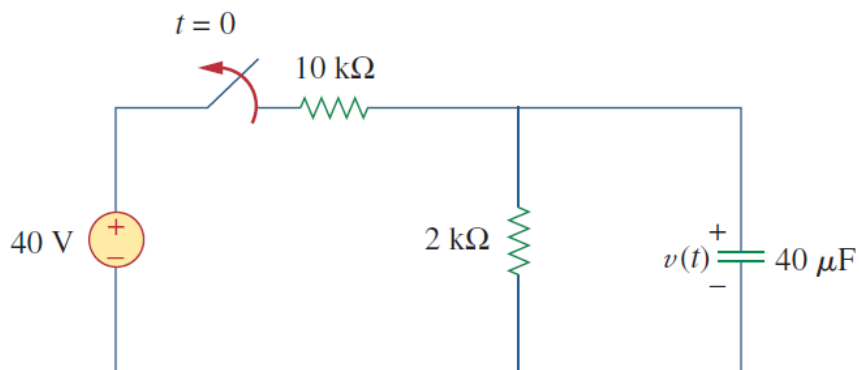
Q: 1

The switch in Fig. has been in position A for a long time. Assume the switch moves instantaneously from A to B at $t = 0$. Find v for $t > 0$.



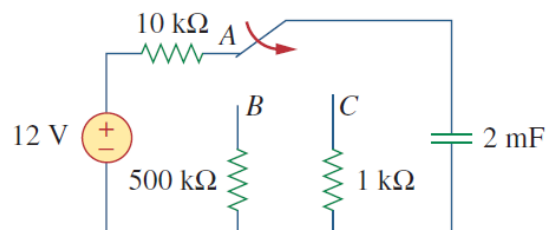
Q2:

The switch in has been closed for a long time, and it opens at $t = 0$. Find $v(t)$ for $t \geq 0$.



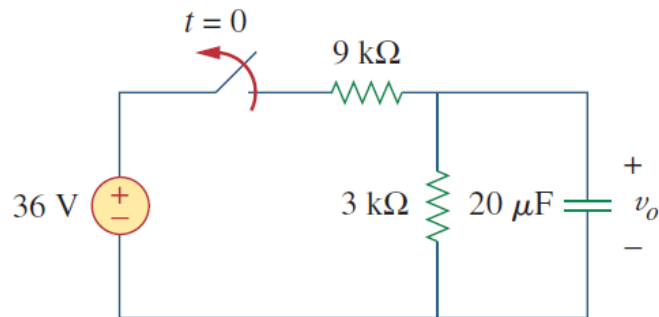
Q:3

Assuming that the switch in ' has been in position A for a long time and is moved to position B at $t = 0$, Then at $t = 1$ second, the switch moves from B to C . Find $v_C(t)$ for $t \geq 0$.

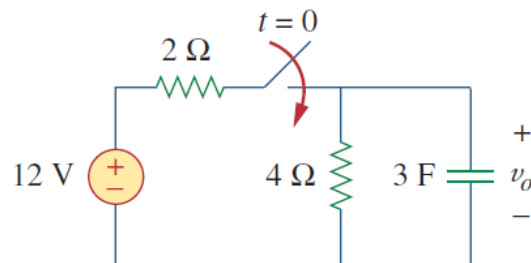


Q:4

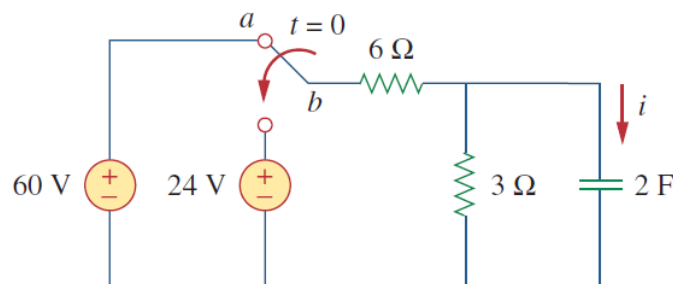
For the circuit in find $v_o(t)$ for $t > 0$.
 Determine the time necessary for the capacitor voltage to decay to one-third of its value at $t = 0$.

**Q:5**

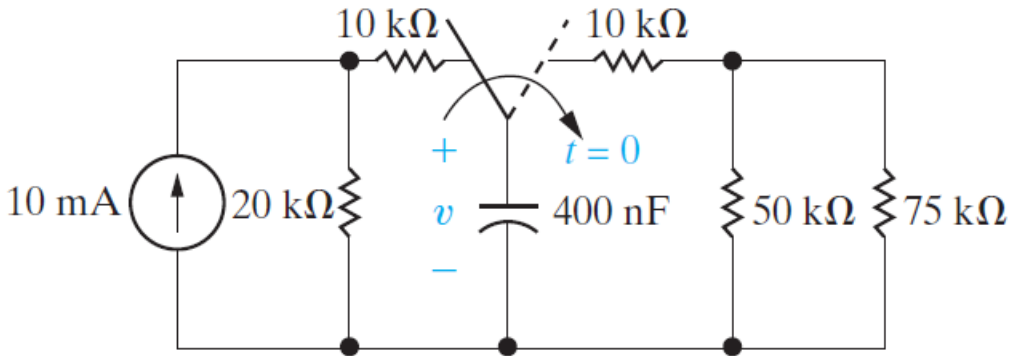
- (a) If the switch in has been open for a long time and is closed at $t = 0$, find $v_o(t)$.
 (b) Suppose that the switch has been closed for a long time and is opened at $t = 0$. Find $v_o(t)$.

**Q:6**

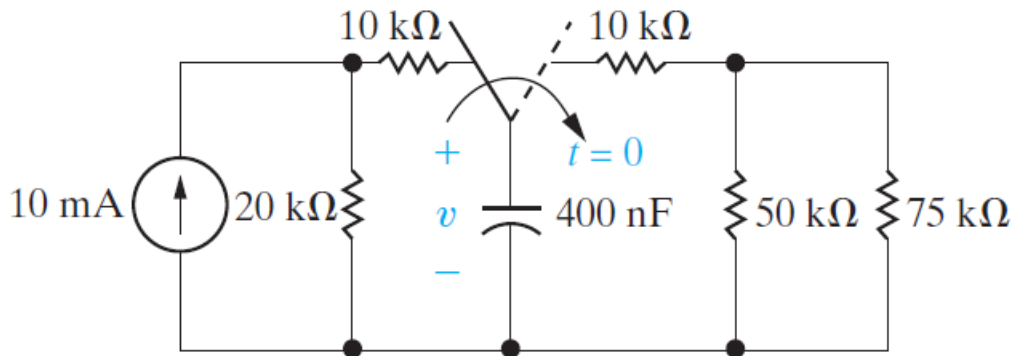
The switch has been in position a for a long time. At $t = 0$, it moves to position b . Calculate $i(t)$ for all $t > 0$.



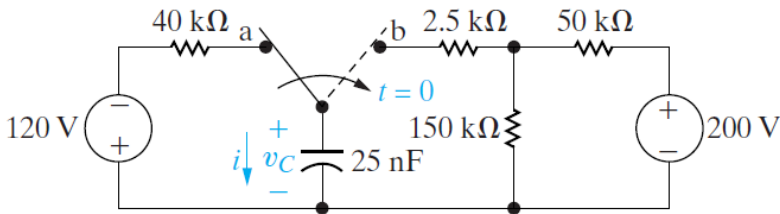
Q:7

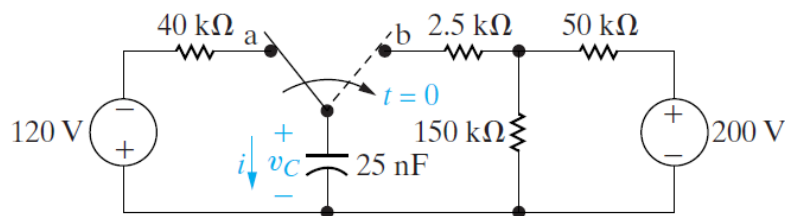
The switch in the circuit in  has been in the left position for a long time. At $t = 0$ it moves to the right position and stays there.

- Find the initial voltage drop across the capacitor.
- Find the initial energy stored by the capacitor.
- Find the time constant of this circuit for $t > 0$.
- Write the expression for the capacitor voltage $v(t)$ for $t \geq 0$.



Q: 8

Assume that the switch in the circuit  has been in position a for a long time and that at $t = 0$ it is moved to position b. Find (a) $v_C(0^+)$; (b) $v_C(\infty)$; (c) τ for $t > 0$; (d) $i(0^+)$; (e) $v_C, t \geq 0$; and (f) $i, t \geq 0^+$.



Q: 9

The switch in the circuit has been in position a for a long time. At $t = 0$, the switch moves instantaneously to position b. For $t \geq 0^+$, find

- a) $v_o(t)$.
- b) $i_o(t)$.

